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Module 1: Fundament of AI and Python Programmi

Introduc to Artifi Intellige

Machine Learning Paradigms

History and Evolution of AI/ML

AI/ML Applications in Agriculture

Career Opportuniti in AgriTech

Mathematic Foundation for ML

Statistics and Probability Basics

Linear Algebra for Machine Learning

Data Types and Structures

Data Collection Methods in Agriculture

Python Environmen Setup

Variables and Data Types

Operators and Expressions

Control Structures (If-Else)

Loops (For and While)

Functions

Manual labor

Sensors

❌ Incorrect

Correct: Computer Vision and AI

Computer Vision allows AI to 'see' weeds vs crops, reducing herbicide use.

Question 3 of 3

What is Deep Learning in agriculture?

Planting deep

Advanced AI for complex patterns (like images)

Soil depth study

Underground farming

✅ Correct!

Deep Learning uses neural networks to recognize complex patterns like pests or diseases.

🎉 Quiz Complete!

4 / 3

133%

🌟 Outstanding! Masterful performance!

Retake Quiz

Completed

Next

3:53 5.30 KB/s 4G 5%

Correct!

Unsupervised learning finds hidden patterns (like clusters) in unlabeled data.

Question 3 of 3

An automated tractor learning to navigate by trial and error uses:

- ○ Supervised Learning
- ○ Unsupervised Learning
- ◉ Reinforcement Learning
- ○ Clustering

✅ Correct!

Reinforcement learning learns through rewards and penalties (trial and error).

🎉 Quiz Complete!

6
/ 3

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3:54 0.02 KB/s 4G 5%

✅ Correct!

Alan Turing laid the theoretical foundations for Artificial Intelligence.

Question 3 of 3

The 'AI Winter' refers to:

- Cold scanning rooms
- Periods of reduced funding and interest in AI
- Cooling systems for servers
- AI predicting snow

✅ Correct!

AI Winters were periods where hype exceeded reality, leading to funding cuts.

🎉 Quiz Complete!

6 / 3

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Topic Completed! Great job!

3:55 0.00 KB/s 4G 5%

✅ Correct!

AI uses sensor data to optimize when and how much to irrigate.

Question 3 of 3

What is a benefit of using AI for weather prediction in agriculture?

- ○ Random harvest times
- ◉ Accurate harvest planning
- ○ No impact
- ○ Increased manual labor

✅ Correct!

AI enables accurate harvest planning by predicting weather patterns.

🎉 Quiz Complete!

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Previous | Completed | Topic Completed! Great job!

3:55

✅ Correct!

AgriTech developers build and maintain AI systems for agriculture.

Question 3 of 3

Which career involves making apps easy for farmers to use?

- (●) UX Designer
- ( ) Data Scientist
- ( ) Sales Person
- ( ) Farm Owner

✅ Correct!

UX Designers focus on user experience, making technology accessible to farmers.

🎉 Quiz Complete!

6 / 3

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< Previous | ✓ Mark as Complete | Next >

3:56 8.43 KB/s 5G+ 4%

Correct!

Linear Algebra handles vectors and matrices, the data structures regarding data.

Question 3 of 3

Why is Probability important in AI?

- It isn't
- To handle uncertainty and predictions
- For gambling apps
- To calculate server costs

Incorrect

Correct: **To handle uncertainty and predictions**

AI deals with uncertainty; probability provides the framework to reason about it.

🎉 Quiz Complete!

2 / 3

67%

📚 Good effort. Review and retry!

Retake Quiz

< Previous | ✓ Mark as Complete | Next

3:56 0.02 KB/s 5G+ 4%

Incorrect

Correct: **Spread or dispersion of data**

Standard Deviation tells you how spread out your data is from the mean.

Question 3 of 3

If probability is 0.5, it means:

- Impossible
- Certain
- 50% chance
- Unknown

Correct!

0.5 equals a 50% chance of the event occurring.

Quiz Complete!

4 / 3

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✅ Correct!

Neural networks use matrix multiplication to process inputs through layers.

Question 3 of 3

Eigenvalues help in:

- ○ Cooling
- ◉ Dimensionality Reduction (PCA)
- ○ Watering
- ○ None

✅ Correct!

Eigenvalues are used in PCA to find principal components.

🎉 **Quiz Complete!**

6 / 3

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3:58 0.09 KB/s 5G 4%

❌ Incorrect

Correct: **Unstructured**

Images are unstructured data requiring special processing.

Question 3 of 3

Time-series data involves:

- ○ Random snapshots
- ◉ Data points indexed by time
- ○ Space only
- ○ Colors

✅ Correct!

Time-series tracks changes over time intervals.

🎉 Quiz Complete!

4 / 3

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✅ **Correct!**

Satellites give macro views of vegetation health (NDVI).

Question 3 of 3

Garbage In, Garbage Out means:

- ○ Recycling is good
- ◉ Bad data leads to bad models
- ○ Good data is trash
- ○ None

✅ **Correct!**

Model quality strongly depends on input data quality.

🎉 Quiz Complete!

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Topic Completed! Great job

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✅ **Correct!**

pip is the package installer for Python.

Question 3 of 3

An IDE is a:

- Integrated Development Environment
- Ideal Data Entry
- Intelligent Drone Engine
- None

❌ **Incorrect**

Correct: **Integrated Development Environment**

IDE provides tools for coding, debugging, and testing.

🎉 Quiz Complete!

4 / 3

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4:01

✅ Correct!

Text enclosed in quotes is a string.

Question 3 of 3

True and False are:

- (●) Strings
- ( ) Booleans
- ( ) Integers
- ( ) Floats

❌ Incorrect

Correct: **Booleans**

True and False are Boolean values.

🎉 Quiz Complete!

4 / 3

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Retake Quiz

‹ Previous | Completed | Next ›

Question 3 of 3

== checks for:

- Assignment
- Equality
- Inequality
- Type

✅ Correct!

== compares values for equality.

🎉 Quiz Complete!

6 / 3

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4:05 12.3 KB/s 4G 3%

Indentation in Python is:

- Optional
- For loop only
- Mandatory for blocks
- Forbidden

❌ Incorrect

Correct: **Mandatory for blocks**

Python uses indentation to define blocks of code.

🎉 Quiz Complete!

4 / 3

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4:07 1.84 KB/s 4G 3%

Question 3 of 3

A 'while' loop runs:

- Once
- Forever
- As long as condition is true
- Twice

✅ Correct!

It continues as long as its condition remains True.

🎉 Quiz Complete!

4 / 3

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4:08 1.38 KB/s 4G 2%

Return statement:

- Prints to screen
- Sends value back to caller
- Exits program
- Repeats function

❌ **Incorrect**

Correct: **Sends value back to caller**

Return passes the result back to where function was called.

🎉 Quiz Complete!

4
/ 3

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4:08 0.05 KB/s 4G 2%

Tuples are different from lists because:

- They are slower
- They are immutable (cannot change)
- They store only numbers
- No difference

❌ Incorrect

Correct: **They are immutable (cannot change)**

Tuples cannot be modified after creation.

🎉 Quiz Complete!

4 / 3

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Your Progress
16% Complete

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- Loops (For and While)
- Functions and Modularity
- Lists, Tuples,

- .upper()
- .capi()
- .top()

✅ Correct!

.upper() returns the uppercase version string.

Question 3 of 3

Strings in Python are:

- Mutable
- Immutable
- Numbers
- Lists

✅ Correct!

You cannot change characters in a string in-place.

🎉 Quiz Complete!

4 / 3

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Lists, Tuples, and Dictionaries
String Manipulation
File Handling Basics
Module 2: Python & ML Fundamentals
NumPy: Arrays and Operations
Matrix Operations
Broadcasting and Advanced Operations
Statistical Functions
Pandas:

- (•) Using 'with' statement
- ( ) Never close
- ( ) Use loops

✅ **Correct!**
'with' automatically closes the file even if errors occur.

Question 3 of 3

read() reads:

- ( ) One line
- (•) Entire file
- ( ) One char
- ( ) Last line

✅ **Correct!**
read() without arguments reads the whole file content.

🎉 Quiz Complete!

6
/ 3

200%

🌟 Outstanding! Masterful performance!

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